

How to Install PostgreSQL for This Course

Two supported methods: Installer or Docker

Choose one — both work for the course.

1. Method 1 — Install PostgreSQL via Installer

This is the simplest option if you want a traditional local installation.

Step 1 — Download PostgreSQL

1. Go to the official PostgreSQL download page.
2. Choose your operating system: Windows, macOS, or Linux.
3. Download the installer provided by EDB.

Step 2 — Run the installer

During installation:

- accept the default settings
- keep the default port **5432**
- set a password for the **postgres** superuser and write it down
- install pgAdmin if you want a GUI (not required for this course)

Step 3 — Verify installation

1. Open a terminal.
2. Run:

```
psql --version
```

If you see a version number, PostgreSQL is installed correctly.

2. Method 2 — Install PostgreSQL via Docker (recommended)

This is the cleanest option: no system-level installation, easy to reset, and identical across machines.

Step 1 — Install Docker Desktop

1. Download Docker Desktop for your operating system.
2. Install and open it.
3. Make sure Docker is running.

Step 2 — Start PostgreSQL using Docker

Run this command from the directory containing **docker-compose.yml**:

```
docker compose -f docker-compose.yml up -d
```

Step 3 — Verify

Run:

```
docker ps
```

You should see a container named **postgres**. You can also verify with Docker Desktop's GUI.

3. Install the VS Code PostgreSQL Extension

Use this extension to run SQL queries directly inside VS Code.

Step 1 — Open VS Code

Step 2 — Install the extension

1. Open the Extensions panel.
2. Search for **PostgreSQL** by **Microsoft**.
3. Click **Install**.

Step 3 — Connect to your database

1. Press **F1**.
2. Type **PostgreSQL: New Connection**.
3. Enter the connection details:

```
Host: localhost  
Port: 5432  
User: postgres  
Password: postgres  
Database: demo
```

You should now be connected.

Step 4 — Run SQL queries

1. Create a file named **queries.sql**.
2. Write SQL inside it.
3. Press **F1**.
4. Type **PostgreSQL: Run Query**.

Results appear inside VS Code.